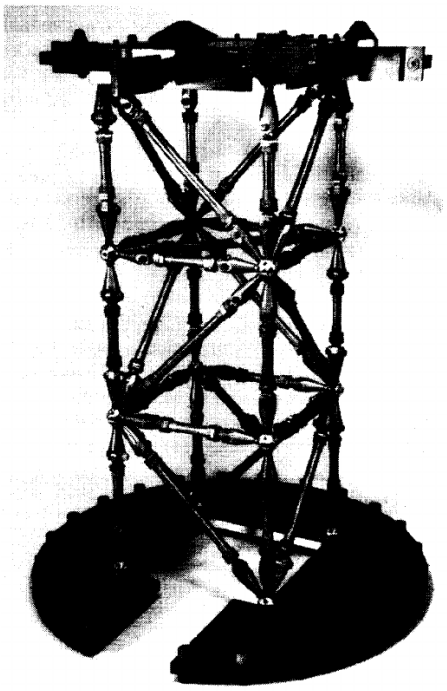
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The damping number that refuses to stay put



A small truss with a large lesson

In September 1995, a pin-jointed truss rode aboard Endeavour on STS-69. The Joint Damping Experiment asked a deceptively simple question: would the structure lose vibration energy in orbit the way it did on the ground? Researchers plucked it and recorded the decay [1].

The resulting report described damping rates changing by factors of three to eight when the truss orientation changed, and damping that depended on amplitude. Reduced gravity-induced preload increased damping in this carefully assembled test structure [1].

Historical photograph: JDX truss, NASA-CR-4781, Figure 1, printed p. 6 (June 1997). Credit: Folkman et al., Utah State University / NASA; public use permitted. Image and report [1]. Scan quality preserved.

That result is a useful interruption to a familiar modeling habit. A damping ratio often enters a finite-element model as one tidy percentage. Yet that number may describe a particular joint state, response level and test setup, rather than a transferable property of the hardware.

Today we will follow the energy behind that percentage, use a decay measurement to expose its assumptions, and see what changes when the value is carried into a resonance prediction. The practical objective is to recognize when a single modal damping value is defensible and when it needs an amplitude range attached.

A joint is a mechanism, not a percentage

For a linear viscous oscillator, damping force is proportional to velocity. Its free-vibration peaks fall exponentially, and one damping ratio describes the whole decay. That is an exceptionally useful model. A contacting interface, however, can stick, slip or separate. Its energy loss need not follow the same rule.

Picture an ideal spring supporting a mass that slides against a constant friction force. Over a complete, symmetric cycle of amplitude A, sliding distance is 4A. The energy dissipated is therefore $4F_cA$, where F_c is the friction-force magnitude. Meanwhile, the spring stores energy proportional to A squared. The numerator and denominator grow at different rates.

For light damping and approximately harmonic motion, an energy-equivalent viscous ratio is

$$\zeta_{eq} \approx \frac{\Delta E}{4\pi E_{max}}$$

Here ΔE is energy lost per cycle [J] and E_{max} is maximum stored elastic energy [J]. For this ideal full-sliding spring system, with stiffness k [N/m] and amplitude A [m], the substitution gives

$$\zeta_{eq} \approx \frac{2F_c}{\pi kA}$$

Doubling amplitude halves this equivalent ratio, even though the friction force has not changed. This is our own idealized energy calculation, not a model fitted to JDX. It applies only while sustained sliding and the near-harmonic approximation remain reasonable. Extrapolating it toward zero amplitude would produce an absurdly large ratio; sticking ends that extrapolation.

Real interfaces can show a different trend. At small excitation, much of a contact may stick; increasing motion can activate more slip and raise dissipation. At larger amplitudes, stored energy may grow faster than loss. There is no universal instruction to increase or decrease damping with level. NASA’s nonlinear joint-modeling work explicitly treats frequency and damping as functions of response amplitude [2].

JDX used clearance-fit pinned joints, so it should not be treated as a universal bolted-joint calibration. Its report identified friction and impacts as important mechanisms and discussed transfer into higher modes [1]. That distinction matters: decay of one measured mode can reflect energy redistribution as well as irreversible dissipation. The measurement and the model must agree on what “loss” means.

The straight line you should look for

A logarithmic-decrement estimate is attractive because it needs little machinery. Take two positive peaks of the same isolated mode, separated by N complete cycles. For an unforced, linear, underdamped single-degree-of-freedom oscillator [3],

$$\delta = \frac{1}{N}\ln\left(\frac{A_0}{A_N}\right)$$

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

A_0 and A_N use the same displacement units; δ and ζ are dimensionless. The following numbers are synthetic local linear examples, deliberately assigned different ratios. They are neither telemetry nor a reconstruction of JDX.

Illustrative decay	A_0 [mm]	A_{10} [mm]	Inferred ζ
Small-motion case	0.1000	0.07304	0.50%
Large-motion case	1.0000	0.28454	2.00%

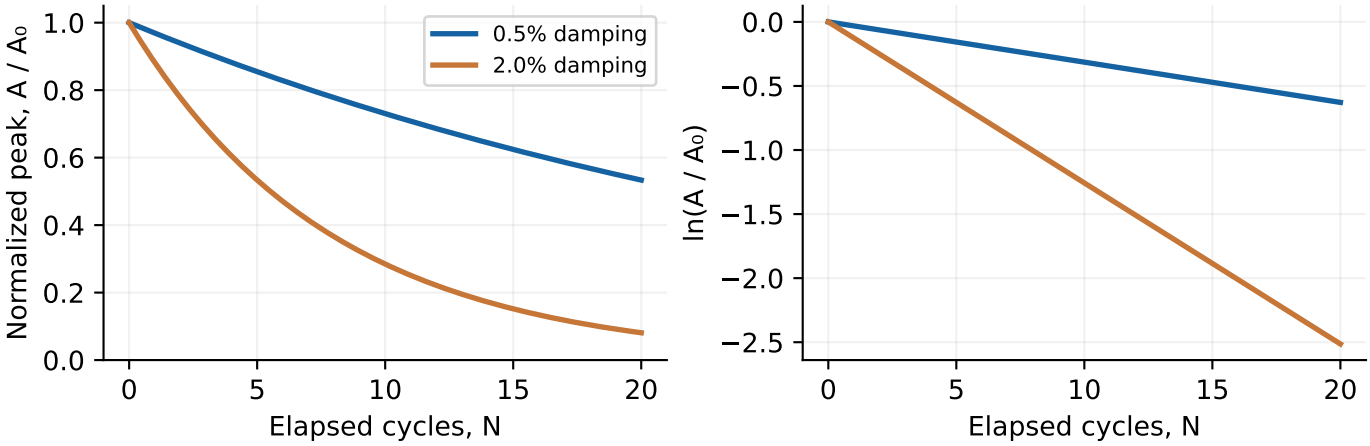


Figure 1. Original analytical envelopes for the two illustrative ratios. Normalization removes the different starting amplitudes. Constant viscous damping gives a straight line on the logarithmic plot. These are independent local models, not one nonlinear decay law.

For the large-motion case, the peak ratio gives δ ≈ 0.12569 per cycle; the equation then returns ζ ≈ 0.0200. The useful diagnostic is the whole envelope. Curvature in its logarithm means a constant decay rate is questionable. Before attributing that curvature to joint friction, check modal beating, residual forcing, filtering and the noise floor. An attractive exponential fit cannot establish the mechanism by itself.

Four times the damping is a big assumption

Suppose an analyst takes the illustrative 2% value from a larger-motion test and uses it to predict a small operational disturbance. If the relevant local ratio is actually 0.5%, the model has imported four times too much damping. To see the consequence cleanly, hold modal mass, stiffness, forcing and frequency fixed.

For a force-driven linear oscillator at its undamped natural frequency, displacement amplitude X is

$$X(\omega_n) = \frac{F_0}{2\zeta k}$$

F_0 is harmonic force amplitude [N], k is stiffness [N/m], and X is displacement amplitude [m]. The corresponding response ratio is

$$\frac{X_{0.5\%}}{X_{2\%}} = \frac{0.020}{0.005} = 4$$

A predicted 0.10 mm becomes 0.40 mm in this sensitivity calculation. That is a fourfold amplitude change without moving the natural frequency at all. A well-correlated frequency, by itself, therefore cannot protect a resonant displacement or stress prediction from an inappropriate damping assumption.

This comparison has an important boundary: both candidate linear models must be used within the amplitude range they represent. If the larger predicted motion changes contact state, the 0.5% model may cease to apply before reaching 0.40 mm. The ratio exposes sensitivity; it is not a self-consistent nonlinear response solution. A level-dependent model must reconcile the predicted amplitude with the damping and stiffness appropriate to that amplitude.

Broadband forcing changes the arithmetic

For an isolated, lightly damped mode under force noise that is approximately white across its resonance, displacement variance scales approximately with the inverse of damping. RMS displacement therefore scales with the inverse square root. Under those assumptions, changing 2% to 0.5% gives about twice the RMS response, rather than four times. This follows from integrating the squared linear frequency-response function over frequency.

A narrow tone samples the resonance height; broadband noise also samples its width. Neither scaling is a universal multiplier for a spacecraft PSD. Closely spaced modes, nonflat excitation, changed modal participation, amplitude-dependent stiffness and nonlinear coupling require the actual response calculation. The engineering question is which part of the transfer function the load occupies.

Put the conditions next to the number

A useful damping result arrives with its mode, response-amplitude range, preload and support configuration, temperature, estimation method and uncertainty. Treat these as part of the result. Recording only “2%” strips away the information needed to decide whether that value belongs in a launch-loads model, a pointing analysis or a component test simulation.

Plan enough excitation levels to look for changes in both decay rate and frequency. Repeating a low-level measurement after a higher-level run also helps separate reversible level dependence from a lasting change in the assembly. Check the response at the joint or a justified proxy: identical shaker acceleration does not guarantee identical local contact motion.

When the evidence supports a linear approximation over the relevant range, use it. When it does not, bound the prediction with physically justified parameter sets or calibrate a nonlinear model against multiple levels. NASA’s quasi-static modal approach offers one route, but its stated assumptions include invariant, uncoupled modes; an arbitrary nonlinear transient requires an additional dynamic model [2]. That is a method with a domain of validity, not a universal qualification requirement.

Three takeaways

1. Damping belongs to a mode and operating state; joints can make it amplitude- and preload-dependent.
2. A decay fit is a model test. Inspect the envelope and competing explanations before assigning a physical mechanism.
3. Check the loading bandwidth and response level before transferring a damping value into a safety or reliability prediction.

Reflection: A high-level test yields more damping than a low-level test. Which value should support a small on-orbit disturbance analysis?
Compact answer: The value validated near the expected operational response and flight joint state, with uncertainty. The high-level result is relevant evidence, but it does not automatically bound small-motion response conservatively.

Sources and scope

[1] Folkman et al., The Joint Damping Experiment (JDX), NASA-CR-4781, June 1997. Summary pp. 1-2; photograph p. 6; flight discussion pp. 41-50. Experimental findings apply to this particular truss.

[2] NASA NESC, Nonlinear Joint Modeling of Complex Systems Using a Quasi-Static Modal Analysis Approach, Sept. 23, 2021. Amplitude dependence and method limitations.

[3] M. J. Casiano, Extracting Damping Ratio From Dynamic Data and Numerical Solutions, NASA/TM-2016-218227, Sept. 2016, §1.3 and equations 17-23. Linear decay assumptions and damping extraction.

Calculations and Figure 1: original educational examples. No program-specific damping values or acceptance criteria are prescribed. Prepared with the available model; GPT-6 Astra selection was unavailable.